

COI - AT1
PSEUDOCODE WRITING

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Java Programming

1) Design a class Hierarchy :-

* Pseudocode :

CLASS Person

DATA

name

age

METHOD read Person()

READ name

READ age

END METHOD

END CLASS.

CLASS Student INHERITS Person

DATA

reg No

marks[5]

average

METHOD readStudent()

READ reg No

FOR $i \leftarrow 0$ TO 4

READ marks[i]

END FOR

END METHOD

CLASS ResearchStudent INHERITS Student

DATA

researchArea

METHOD readResearch()

READ researchArea

END METHOD

METHOD - display complete details ()

PRINT name

PRINT age

PRINT reg No

PRINT average

PRINT researchArea

END METHOD

END CLASS

BEGIN

CREATE obj AS Research Student

obj. read Person()

obj. read Student()

obj. read Research()

obj. calculateAverage()

obj. display complete details()

END.

2) Complete the Missing PseudoCode :-

BEGIN

READ N

READ ARRAY A

FOR $i \leftarrow 0$ TO $N-1$

FOR $j \leftarrow i+1$ TO $N-1$

IF $A[i] == A[j]$

DELETE $A[j]$

END IF

END FOR

END FOR

PRINT MODIFIED ARRAY

END.

3)

Debug the Pseudocode :-

```
CLASS Area
METHOD area (side)
RETURN side * side
END METHOD
METHOD area (length, breadth)
RETURN length * breadth
END METHOD
END CLASS.
```

4)

Trace the Execution :-

* Pseudo :

sum = 0

1 odd $\rightarrow -1$

2 even $\rightarrow +2$

3 odd $\rightarrow -3$

4 even $\rightarrow +4$

5 odd $\rightarrow 5$

* calculation :

$0 - 1 = -1$

$-1 + 2 = 1$

$1 - 3 = -2$

$-2 + 4 = 2$

$2 - 5 = -3$

* Output :

-3.

5) Write Efficient Pseudocode :-

```
BEGIN
READ N
READ first element
largest ← first element
second largest ← -∞
FOR i ← 2 TO N
    READ x
    IF x > largest THEN
        second largest ← largest
        largest ← x
    ELSE IF x > second largest AND x ≠ largest THEN
        second largest ← x
    END IF
END FOR
PRINT second largest
END.
```

6) Convert Problem Statement into Pseudocode :-

```
INTERFACE Payment
    METHOD pay (amount)
END INTERFACE

CLASS CreditCard IMPLEMENTS Payment
    METHOD pay (amount)
        PRINT "Paid", amount, "using credit card"
    END METHOD
END CLASS

CLASS UPI IMPLEMENTS Payment
```

```

METHOD pay (amount)
    PRINT "Paid", amount, "using UPI"
END METHOD
END CLASS .

CLASS NetBanking IMPLEMENTS Payment
    METHOD pay (amount)
        PRINT "Paid", amount, "using Net Banking"
    END METHOD
END CLASS

BEGIN
    READ choice
    READ amount
    DECLARE Payment P
    IF choice = 1 THEN
        P ← NEW creditCard
    ELSE IF choice = 2 THEN
        P ← NEW UPI
    ELSE
        P ← NEW NetBanking
    END IF
    P.pay (amount)
END .

```

7) complete the Pseudocode :-

```

CLASS Animal
    METHOD speak()
        PRINT "Animal"
    END METHOD

```



```

END CLASS
CLASS Dog EXTENDS Animal
METHOD speak()
    PRINT "Barks"
END METHOD
END CLASS
Animal obj ← NEW Dog
obj.speak().

```

8)

Find Logical Errors :-

```

CLASS Employee
DATA
    id
    name
CONSTRUCTOR (id, name)
    this.id ← id
    this.name ← name
END CONSTRUCTOR
METHOD display()
    PRINT "Employee ID :", id
    PRINT "Employee Name:", name
END METHOD
END CLASS
BEGIN
    CREATE emp AS Employee (101, "Hema Puriya")
    emp.display()
END .

```

9)

Case Study :-

* Pseudocode :

CLASS Vehicle

DATA

Vehicle No

Owner Name

METHOD read()

READ Vehicle No

READ Owner Name

END METHOD

METHOD display()

PRINT Vehicle No

PRINT Owner Name

END METHOD

END CLASS

CLASS Car INHERITS Vehicle

DATA

Fuel Type

METHOD read()

CALL Vehicle.read()

READ fuel Type

END METHOD

METHOD display()

CALL Vehicle.display()

PRINT fuel Type

END CLASS

CLASS Truck INHERITS Vehicle

DATA

load Capacity

METHOD read()

CALL vehicle.read()

READ load Capacity

END METHOD.

METHOD display()

CALL vehicle.display()

PRINT load Capacity

END METHOD

END CLASS

BEGIN

READ N

DECLARE vehicle list [N]

FOR $i \leftarrow 0$ TO $N-1$

READ Type

IF Typ = "Car" THEN

list[i] $\leftarrow$ NEW Car

ELSE

list[i] $\leftarrow$ NEW Truck

END IF

list[i].read()

END FOR

FOR $i \leftarrow 0$ TO $N-1$

list[i].display()

END FOR

END.